

Solvable
v
Constructing nonabelian extensions

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...from Galois theory class: Abelian examples

ζ_7 is a 7th root of unity. e.g. $\zeta_7 = e^{\frac{2\pi i}{7}}$.

$\mathbb{Q}(\zeta_7)$

satisfies $\frac{x^7-1}{x-1} = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$. ✓

the 7th roots of unity are $\zeta_7^i : 1 \leq i \leq 6$.

$\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) = \{\sigma_i : 1 \leq i \leq 6\}$ with $\sigma_i(\zeta_7) = \zeta_7^i$. ✓

6

These satisfy $\sigma_i \circ \sigma_j(\zeta_7) = \sigma_i(\zeta_7^j) = (\sigma_i(\zeta_7))^j = \zeta_7^{ij}$.

Group isomorphism exists between $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ and the multiplicative group of the integers modulo 7.

$\mathbb{Q}$

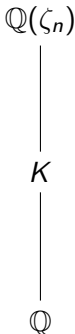
This is a cyclic group of order 6 generated by 3, since

$$3, 3^2 \equiv 2, 3^3 \equiv 6, 3^4 \equiv 4, 3^5 \equiv 5, 3^6 \equiv 1 \pmod{7}.$$

As you are learning this week...

Kronecker-Weber Theorem:

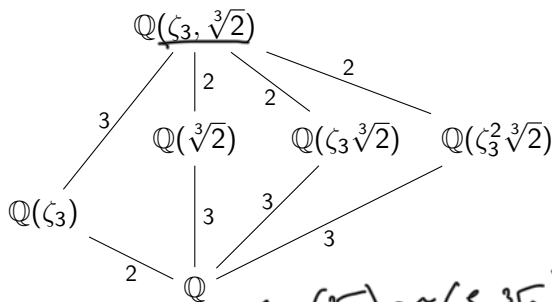
Given any Galois extension $K/\mathbb{Q}$ with $\text{Gal}(K/\mathbb{Q})$ abelian,
there exists an integer n such that $K \subseteq \mathbb{Q}(\zeta_n)$.



The class examples (cyclotomic examples) you saw are a good representation of what you might see in any abelian extension.

On the other hand... nonabelian examples

In this case, we define two ring isomorphisms from $\mathbb{Q}(\zeta_3, \sqrt[3]{2})$ to itself:



σ determined by

$$\sigma(\zeta_3) = \zeta_3,$$

$$\sigma(\sqrt[3]{2}) = \zeta_3\sqrt[3]{2}.$$

τ determined by

$$\tau(\zeta_3) = \zeta_3^2 \text{ and}$$

$$\tau(\sqrt[3]{2}) = \sqrt[3]{2}.$$

$$\tau(\sqrt[3]{2}) = \tau(\zeta_3\sqrt[3]{2}) = \zeta_3\sqrt[3]{2}$$

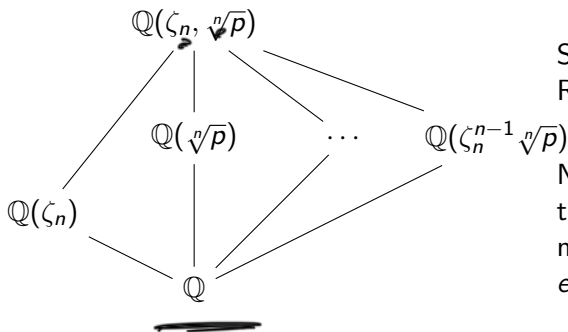
$$\tau^2(\sqrt[3]{2}) = \tau^2(\sqrt[3]{2}) = \zeta_3^2\sqrt[3]{2} //$$

$$\langle \sigma, \tau : \sigma^3 = \tau^2 = 1, \tau\sigma = \sigma^2\tau \rangle \cong D_3$$

$D_3 \cong$ the group of invertible linear functions $f_{a,b}(x) = ax + b$ modulo 3.

Alternatively, the group of invertible matrices $\begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$ modulo 3.

Typical class examples generalize this: $\mathbb{Q}(\zeta_n, \sqrt[n]{p})$, p prime

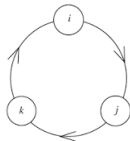


Similar Galois group.
Replace "3" with "n".

Note $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong U(n)$,
the units modulo n , which
may not be cyclic.
e.g. $U(8)$

But what about other nonabelian groups? Such as...

the group of quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$?



or for any $p > 3$, the Heisenberg group modulo p ?

$$H(p) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \bmod p \right\}$$

Our mindset/orientation today

Start with an abstract p -group. Construct an extension.

In fact, construct all possible extensions that have this Galois group.

Be explicit. Produce equations.

Outline

I. Transition to characteristic p . Notation. Galois theory facts.

II. Normal Basis Theorem $\implies$ *Additive* Hilbert Theorem 90



$\implies$ Cyclic degree p extensions are Artin-Schreier.

III. Construct Heisenberg extensions.

IV. Then out of the “blue”: the Algebraic group approach.

Transition

Let p be prime,

$\mathbb{F}_p$ be the field with p elements (the integers modulo p).

The polynomial ring $\mathbb{F}_p[t]$ shares many properties with $\mathbb{Z}$.

Its quotient field, the field of rational functions

$$\mathbb{F}_p(t) = \left\{ \frac{r(t)}{s(t)} : r(t), s(t) \in \mathbb{F}_p[t], s(t) \neq 0 \right\}$$

shares properties with $\mathbb{Q}$,

and has for a long time been a source of insight and inspiration.

Some Galois theory facts

Galois extensions are separable and normal.

- ① Separable: We've avoided the problem with the irreducible polynomial $x^p - t \in K[x]$ where $K = \mathbb{F}_p(t)$. Problem: Let α be a root. But then $x^p - t = x^p - \alpha^p = (x - \alpha)^p$ has only one root!

We can “separate” the roots of the defining polynomial.

There is a nontrivial field isomorphism that maps one root to the other.

- ② Normal: We've avoided the problem with $\mathbb{Q}(\sqrt[3]{2})$. This extension doesn't contain the other two roots of $x^3 - 2$. There are isomorphisms from $\mathbb{Q}(\sqrt[3]{2})$ to other fields. e.g. $\mathbb{Q}(\zeta_3 \sqrt[3]{2})$

Isomorphisms are automorphisms.

Main Theorem of Galois Theory

Let L be a finite Galois extension of K with Galois group $\text{Gal}(L/K)$.

① $|\text{Gal}(L/K)| = [L : K]$. ✓

② Bijective correspondence between subgroups of $\text{Gal}(L/K)$ and intermediate fields M , $K \subseteq M \subseteq L$. ✓

✦ **Normal Basis Theorem:** Let $\text{Gal}(L/K) = \{\sigma_1, \sigma_2, \dots, \sigma_n\}$. There is an $\alpha \in L$ such that

$$\sigma_1(\alpha), \sigma_2(\alpha), \dots, \sigma_n(\alpha)$$

is a basis for L over K .

Examples

- $\mathbb{Q}(\sqrt{2})$: The usual power basis is $\{1, \sqrt{2}\}$ for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$.

A normal basis is $\{1 + \sqrt{2}, 1 - \sqrt{2}\}$. (Check linear independence.)

$$\begin{aligned} a(1 + \sqrt{2}) + b(1 - \sqrt{2}) &= 0 \\ (a + b) + (a - b)\sqrt{2} &= 0 \end{aligned} \quad \begin{aligned} a + b &= 0 \\ a - b &= 0 \\ 2a &= 0 \Rightarrow a = 0 \end{aligned}$$

- $\mathbb{Q}(\zeta_5)$ where ζ_5 is a root of $x^4 + x^3 + x^2 + x + 1 = 0$.

The usual power basis is $\{1, \zeta_5, \zeta_5^2, \zeta_5^3\}$ for $\mathbb{Q}(\zeta_5)$ over $\mathbb{Q}$.

A normal basis is $\{1 + \zeta_5, 1 + \zeta_5^2, 1 + \zeta_5^3, 1 + \zeta_5^4\}$.
(Again, check linear independence.)

Artin-Schreier extensions

Let $K = \mathbb{F}_p(t)$, and suppose $x^p - x = \beta \in K$ is irreducible with root α .

Then $\alpha + 1$ is another root. Indeed, $\{\alpha + i : i \in \mathbb{F}_p\}$ is the set of roots.

$$\begin{aligned}\alpha^p - \alpha &= \beta \\ (\alpha+1)^p - (\alpha+1) &= \alpha^p + 1 - \alpha - 1 = \alpha^p - \alpha = \beta \\ (\alpha+i)^p - (\alpha+i) &= \alpha^p + i^p - \alpha - i = \beta \\ i^p &= i\end{aligned}$$

$K(\alpha)/K$ is Galois with $\text{Gal}(K(\alpha)/K) = \langle \sigma \rangle \cong C_p$, $\sigma(\alpha) = \alpha + 1$.

We want to prove the converse.

Stepping stone: Additive Hilbert Theorem 90

Let L/K be a Galois extension with $\text{Gal}(L/K) \cong C_p$ generated by σ .

Thus $\sigma^p = 1$, the identity.

The trace $\text{Tr} = 1 + \sigma + \sigma^2 + \cdots + \sigma^{p-1}$ is a K -linear map that we can apply to elements of L : Given $\mu \in L$, $\text{Tr}(\mu) = \mu + \sigma(\mu) + \cdots + \sigma^{p-1}(\mu)$.

Suppose $\ell \in L$ with $\text{Tr}(\ell) = 0$. There is an $\alpha \in L$ such that $\ell = \sigma(\alpha) - \alpha$.

We think of $\sigma(\alpha) - \alpha$ as the result of the K -linear map $(\sigma - 1)$ acting on the "vector" α , and write $\ell = (\sigma - 1)\alpha$.

First, let's verify: If $\ell = (\sigma - 1)\alpha$, then $\text{Tr}(\ell) = 0$.

$$\begin{aligned}\ell &= \sigma(\alpha) - \alpha \\ \text{Tr}(\ell) &= \text{Tr}(\sigma(\alpha)) - \text{Tr}(\alpha) \\ &= (\cancel{\sigma(\alpha)} + \sigma^2(\alpha) + \cdots + \sigma^p(\alpha)) - (\cancel{\alpha} + \cancel{\sigma(\alpha)} + \cdots + \cancel{\sigma^{p-1}(\alpha)})\end{aligned}$$

Proof of Additive Hilbert Theorem 90

Preparation: Using the Normal Basis Theorem, there is an $\mu \in L$ such that

$$\{\mu, \sigma(\mu), \sigma^2(\mu), \dots, \sigma^{p-1}(\mu)\}$$

is a K -basis for L .

Notice that: $\text{Tr}(\mu) = \mu + \sigma(\mu) + \dots + \sigma^{p-1}(\mu) \neq 0$. ✓

Change basis: For $1 \leq i \leq p-1$, replace $\sigma^i(\mu)$ with $\sigma^i(\mu) - \mu$.

$$\text{Adjust notation: } \sigma^i(\mu) - \mu = (\sigma^i - 1)\mu.$$

Think: $(\sigma^i - 1)$ is a K -linear function acting on the vector μ .

New basis: $\{\mu, (\sigma - 1)\mu, (\sigma^2 - 1)\mu, \dots, (\sigma^{p-1} - 1)\mu\}$.

Notice that: $\text{Tr}((\sigma^i - 1)\mu) = 0$ for $1 \leq i \leq p-1$, because

$$\{\sigma^i(\mu), \sigma^{i+1}(\mu), \sigma^{i+2}(\mu), \dots, \sigma^{i+p-1}(\mu)\} = \{\mu, \sigma(\mu), \sigma^2(\mu), \dots, \sigma^{p-1}(\mu)\},$$

and thus $\text{Tr}(\sigma^i(\mu)) = \text{Tr}(\mu)$.

We are ready!

Suppose $\ell \in L$ and $\text{Tr}(\ell) = 0$. Since $\ell \in L$, there are $a_i \in K$ such that

$$\ell = a_0\mu + a_1(\sigma - 1)\mu + a_2(\sigma^2 - 1)\mu + \cdots + a_{n-1}(\sigma^{n-1} - 1)\mu.$$

Apply the trace to both sides.

Since $\text{Tr}(\ell) = 0$ and $\text{Tr}((\sigma^i - 1)\mu) = 0$ for $1 \leq i \leq p - 1$,

$$0 = \text{Tr}(a_0\mu) = a_0\text{Tr}(\mu) \text{ with } \text{Tr}(\mu) \neq 0.$$

This means $a_0 = 0$.

And thus that ℓ is a sum of terms $a_i(\sigma^i - 1)\mu$, $i \geq 1$.

Factor $(\sigma - 1)$ out of each term.

We have proven $\ell = (\sigma - 1)\alpha$ for some $\alpha \in L$. Done!

Proving C_p -extensions are Artin-Schreier

Suppose L/K is Galois with $\text{Gal}(L/K) \cong C_p$ generated by σ . ✓

Since the characteristic of $K = \mathbb{F}_p(t)$ is p ,

$$\text{Tr}(1) = 1 + \sigma(1) + \cdots + \sigma^{p-1}(1) = p = 0. \quad \checkmark \quad \sigma(\alpha) = \alpha + 1$$

Using additive Hilbert Thm 90, there is an $\alpha \in L$ such that $1 = (\sigma - 1)\alpha$.

Rewrite as $\sigma(\alpha) = \alpha + 1$. Since $\sigma(\alpha) \neq \alpha$, $K \subsetneq K(\alpha) \subseteq L$ where because $[L : K] = p$, we have $L = K(\alpha)$. Induction yields $\sigma^i(\alpha) = \alpha + i$.

Recall Fermat's little Theorem: $0, 1, \dots, p-1$ solve $x^p \equiv x \pmod{p}$.

So $x^p - x$ factors as $\prod_{i \in \mathbb{F}_p} (x - i)$, which means that

$$\alpha^p - \alpha = \prod_{i \in \mathbb{F}_p} (\alpha - i) = \prod_{i=0}^{p-1} \sigma^i(\alpha) \in K.$$

i.e. α is a root of the Artin-Schreier equation $x^p - x = \beta$ for some $\beta \in K$.

We are ready to start constructing Heisenberg extensions.

actually... Galois extensions for any p -group.

Start with the group

$$H(p): \quad \sigma_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

These elements satisfy $\sigma_i^p = 1$. The center is $\langle \sigma_3 \rangle$. And $\sigma_2 \sigma_1 = \sigma_1 \sigma_2 \sigma_3$.

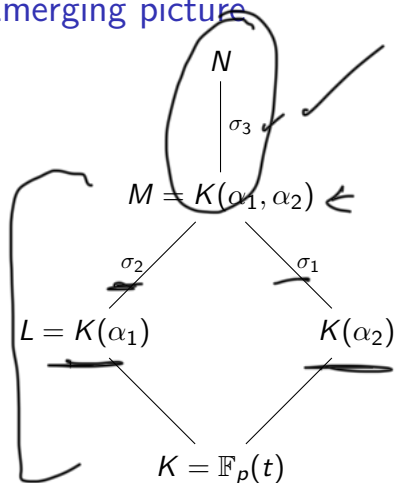
Importantly, $H(p)/\langle \sigma_3 \rangle \cong C_p \times C_p$ is elementary abelian, and we know how elementary abelian extensions are constructed!

Begin: Suppose N/K is a Galois extension with $\text{Gal}(N/K) = \langle \sigma_1, \sigma_2, \sigma_3 \rangle$, and let $M = N^{\sigma_3}$ be the fixed field of $\langle \sigma_3 \rangle$.

Then since $\text{Gal}(M/K) = \frac{\langle \sigma_1, \sigma_2, \sigma_3 \rangle}{\langle \sigma_3 \rangle} \cong C_p \times C_p$,

the extension M/K is built out of two Artin-Schreier C_p -extensions.

Emerging picture



$M = K(\alpha_1, \alpha_2)$ with

α_i satisfying $x^p - x = \beta_i \in K$,

and $\sigma_i(\alpha_j) = \alpha_j + \delta_{ij}$, $i, j \in \{1, 2\}$,

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Kronecker delta.

Since N/M is a C_p -extension, $N = M(\alpha_3)$ for some $\alpha_3^p - \alpha_3 = \mu \in M$.

But M has lots of C_p -extensions. When do we have a $H(p)$ -extension?

This is an *embedding problem*.

Recall: α_1, α_2 satisfying $\sigma_3(\alpha_j) = \alpha_j$, solving $x^p - x = \beta_j \in K$. And

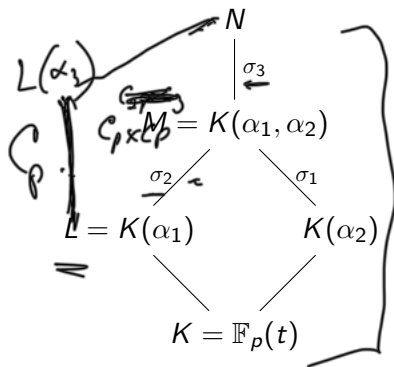
$$\sigma_i(\alpha_j) = \alpha_j + \delta_{ij} \text{ for } i, j \in \{1, 2\}.$$

- Now since σ_3 is in the center, $\langle \sigma_2, \sigma_3 \rangle \cong C_p \times C_p$. Thus N is the composite of two Artin-Schreier extensions of L . Namely, $N = L(\alpha_2, \alpha_3)$ with $\alpha_3^p - \alpha_3 \in L$ and $\sigma_2(\alpha_3) = \alpha_3$

In summary,

$$\sigma_i(\alpha_j) = \alpha_j + \delta_{ij} \text{ for } i, j \in \{1, 2, 3\},$$

except one pair: $(i, j) = (1, 3)$.



For that one pair, all we can say is that $\sigma_1(\alpha_3) \in N$.

So, $\sigma_1(\alpha_3) - \alpha_3 = A$,
for some $A \in N$.

Goal: Determine A .

Determine A

Recall $A = (\sigma_1 - 1)\alpha_3 \in N = K(\alpha_1, \alpha_2, \alpha_3)$.

- I. Since $(\sigma_3 - 1)\alpha_3 = 1$ and the K -linear maps $(\sigma_1 - 1)$ and $(\sigma_3 - 1)$ commute, $(\sigma_3 - 1)A = (\sigma_1 - 1)(\sigma_3 - 1)\alpha_3 = 0$.

$$\sigma_3(A) = A \quad (\sigma_1 - 1)(\alpha_3) = 0 \quad \text{Thus } A \in M = K(\alpha_1, \alpha_2).$$

- II. Use $\sigma_2\sigma_1 = \sigma_1\sigma_2\sigma_3$. Since

$$\begin{aligned}\sigma_2\sigma_1(\alpha_3) &= \sigma_2(\alpha_3 + A) = \alpha_3 + \sigma_2(A), \text{ and} \\ \sigma_1\sigma_2\sigma_3(\alpha_3) &= \sigma_1\sigma_2(\alpha_3 + 1) = \sigma_1(\alpha_3 + 1) = \alpha_3 + A + 1,\end{aligned}$$

We find that $\sigma_2(A) = A + 1$.

Since $A \in M$ and $\sigma_2(\alpha_2) = \alpha_2 + 1$, we find that σ_2 fixes $A - \alpha_2 \in M$.

Thus $A - \alpha_2 \in L = K(\alpha_1)$.

Determining A

We have shown $A = (\sigma_1 - 1)\alpha_3$ satisfies $A - \alpha_2 \in L = K(\alpha_1)$.

III. Using $\sigma_1(\alpha_3) = \alpha_3 + A$, we find

$$\sigma_1(\alpha_3) = \alpha_3 + A,$$

$$\sigma_1^2(\alpha_3) = \alpha_3 + A + \sigma_1(A),$$

$$\sigma_1^3(\alpha_3) = \alpha_3 + A + \sigma_1(A) + \sigma_1^2(A),$$

$\vdots$

$$\sigma_1^p(\alpha_3) = \alpha_3 + A + \sigma_1(A) + \sigma_1^2(A) + \cdots + \sigma_1^{p-1}(A).$$

Thus $\text{Tr}_{\langle \sigma_1 \rangle}(A) = 0$.

Meanwhile σ_1 fixes α_2 . Thus $\text{Tr}_{\langle \sigma_1 \rangle}(\alpha_2) = p = 0$ as well.

Therefore $A - \alpha_2 \in L = K(\alpha_1)$ and $\text{Tr}_{\langle \sigma_1 \rangle}(A - \alpha_2) = 0$. Additive Hilbert Theorem 90 kicks in and there is an $\ell \in L$ such that

$$A - \alpha_2 = (\sigma_1 - 1)\ell.$$

Determining A

Recall that $L(\alpha_3)/L$ is a C_p -extension, that $\alpha_3^p - \alpha_3 \in L$ and that $(\sigma_3 - 1)\alpha_3 = 1$, $(\sigma_2 - 1)\alpha_3 = 0$.

Furthermore, $A = (\sigma_1 - 1)\alpha_3$ and $A - \alpha_2 = (\sigma_1 - 1)\ell$ with $\ell \in L$.

Set $\alpha'_3 = \alpha_3 - \ell$. Then since L is fixed by σ_2, σ_3 , we have $(\sigma_3 - 1)\alpha'_3 = 1$, $(\sigma_2 - 1)\alpha'_3 = 0$. Additionally,

$$(\sigma_1 - 1)\alpha'_3 = (\sigma_1 - 1)(\alpha_3 - \ell) = A - (\sigma_1 - 1)\ell = \alpha_2.$$

So, we replace α_3 with α'_3 and then relabel.

Without loss of generality, $(\sigma_1 - 1)\alpha_3 = \alpha_2$.

What does this tell us about the element $\mu \in L$ such that $\alpha_3^p - \alpha_3 = \mu$?

$$\sigma_1(\mu) = (\sigma_1(\alpha_3))^p - \sigma_1(\alpha_3) = (\alpha_3 + \alpha_2)^p - (\alpha_3 + \alpha_2) = \mu + \beta_2$$

Determining $\mu = \alpha_3^p - \alpha_3$

We have $\sigma_1(\mu) = \mu + \beta_2$. Thus $(\sigma_1 - 1)\mu = \beta_2$.

Since $(\sigma_1 - 1)(\beta_2\alpha_1) = \beta_2$ as well.

We find that because $\mu - \beta_2\alpha_1 \in L = K(\alpha_1)$ is fixed by σ_1 ,

$$\mu - \beta_2\alpha_1 \in K.$$

In other words,

$$\alpha_3^p - \alpha_3 = \alpha_1\beta_2 + \beta_3 \text{ for some } \beta_3 \in K.$$

Summary

Let $K = \mathbb{F}_p(t)$. If N/K is a Galois extension with $\text{Gal}(N/K) \cong H(p)$, then $N = K(\alpha_1, \alpha_2, \alpha_3)$ with

$$\alpha_1^p - \alpha_1 = \beta_1,$$

$$\alpha_2^p - \alpha_2 = \beta_2,$$

$$\alpha_3^p - \alpha_3 = \alpha_1\beta_2 + \beta_3.$$

And the converse holds, assuming the polynomial equations are irreducible.

Quaternion answer.

Quaternion extensions of $K = \mathbb{F}_2(t)$: $N = K(\alpha_1, \alpha_2, \alpha_3)$ with

$$\alpha_1^2 - \alpha_1 = \beta_1,$$

$$\alpha_2^2 - \alpha_2 = \beta_2,$$

$$\alpha_3^2 - \alpha_3 = \alpha_1\beta_1 + \alpha_1\beta_2 + \alpha_2\beta_2 + \beta_3.$$

For more exercises and an answer-key!

Grant Moles, *Non-Abelian extensions of degree p^3 and p^4 in characteristic $p > 2$* . Finite Fields Appl. **96** (2024).

Algebraic group approach

Interestingly (and for me, surprisingly!), the Heisenberg system of equations

$$\left. \begin{aligned}\alpha_1^p - \alpha_1 &= \beta_1, \\ \alpha_2^p - \alpha_2 &= \beta_2, \\ \alpha_3^p - \alpha_3 &= \alpha_1\beta_2 + \beta_3.\end{aligned}\right\}$$

amounts to the single matrix equation

$$\begin{pmatrix} 1 & \alpha_1^p & \alpha_3^p \\ 0 & 1 & \alpha_2^p \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \alpha_1 & \alpha_3 \\ 0 & 1 & \alpha_2 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & \beta_1 & \beta_3 \\ 0 & 1 & \beta_2 \\ 0 & 0 & 1 \end{pmatrix}$$


$H(R) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in R \right\}$ is a group regardless of ring R .

In this talk, $\text{char}(R) = p$.

Since $\alpha_1, \alpha_2, \alpha_3 \in N$, the that matrix equation exists in the group $H(N)$.

The Galois group is the $\mathbb{F}_p$ -points $H(\mathbb{F}_p)$ of this large group.

Galois action is described with a matrix equation (omitted).

The point is that this is a really nice set-up:

Whenever a Galois group is the $\mathbb{F}_p$ -points of such an *Algebraic group*, this sort of thing works.

(Pointed out to me by Kevin Keating, Univ Florida)

Down a rabbit-hole...

You might now wonder whether all p -groups can be realized as the $\mathbb{F}_p$ -points of an algebraic group. And you aren't the first.

David Speyer (Univ Michigan) believes D_8 (order 16) to be the smallest counter-example

mathoverflow.net/questions/262745/
9 years. Still open.

Of course...

I determined the Artin-Schreier equations that produce D_8 -extensions in characteristic 2.

Yes, reverse engineering Keating's observation, yields a binary operation:

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} * \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_0 + y_0 \\ x_1 + y_1 \\ x_2 + y_2 + x_0 y_1 + x_1 y_1 \\ x_3 + y_3 + D + E + F \end{bmatrix}$$

where $D = x_1^3 y_1 + x_1 y_1^3 + x_1 x_2 y_1 + x_1 y_1 y_2 + x_2 y_2$, $E = x_0 x_1 y_1^2 + x_0 x_2 y_1$, and $F = x_0(y_1 + y_2 + y_1 y_2 + y_0 y_1^2)$.

But no, this doesn't produce an algebraic group
(Group operation defined by polynomials).

It's not associative!

Thanks!